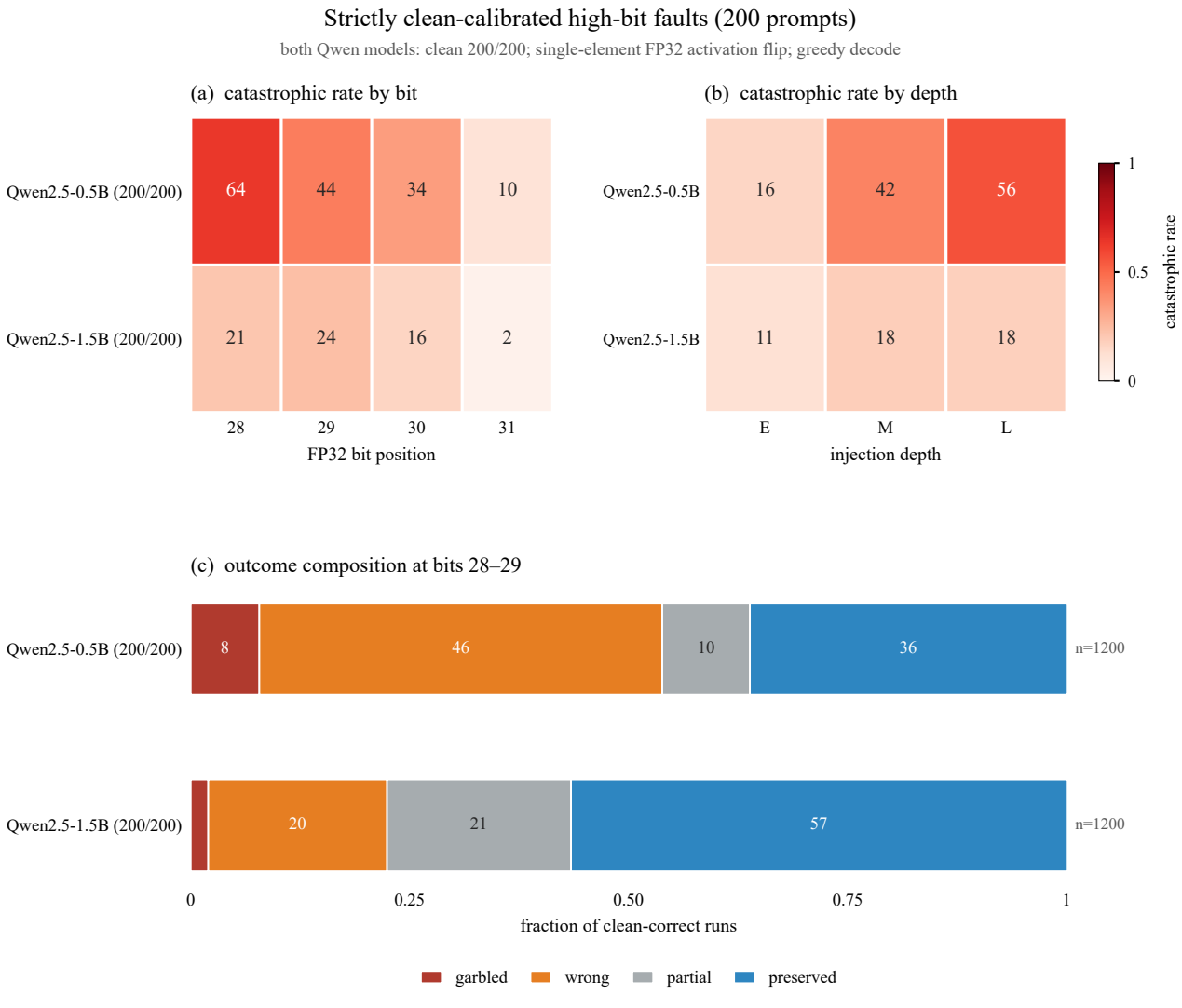
Strictly clean-calibrated high-bit faults (200 prompts)
both Qwen models: clean 200/200; single-element FP32 activation flip; greedy decode
(a) catastrophic rate by bit
Qwen2.5-0.5B (200/200)
Qwen2.5-1.5B (200/200)
64
44
34
10
21
24
16
2
28
29
30
31
FP32 bit position
(b) catastrophic rate by depth
Qwen2.5-0.5B
Qwen2.5-1.5B
16
42
56
11
18
18
E
M
L
injection depth
1
0.5
0
catastrophic rate
(c) outcome composition at bits 28–29
Qwen2.5-0.5B (200/200)
Qwen2.5-1.5B (200/200)
8
46
10
36
20
21
57
n=1200
n=1200
0
0.25
0.50
0.75
1
fraction of clean-correct runs
garbled
wrong
partial
preserved
catastrophic = literal garble or answer absent; partial = answer present but altered; 4,800 fault runs; no NaN/Inf logits.